WORKED SOLUTION

STEP I 1996, Question 1

Pure • Optimisation of Cylinder

STEP I 1996, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q1

The key idea

Two quantities describe the tin, its radius r and its height h. The volume constraint lets us remove one of them, so S becomes a function of r alone, and we can minimise it by ordinary differentiation. The optimal tin turns out to have a very tidy shape. Once you see that shape, both sphere questions come down to looking at a single cross-section through the axis.

Solution

Minimising the surface area

Let the tin have radius $r > 0$ and height $h > 0$. Then

$$V = \pi r^2 h, \qquad S = 2\pi r^2 + 2\pi r h,$$

where $2\pi r^2$ comes from the two ends and $2\pi rh$ from the curved side. For fixed V we have $h = V/(\pi r^2)$, so $2\pi rh = 2V/r$ and

$$S(r) = 2\pi r^2 + \frac{2V}{r}, \qquad r > 0.$$

Differentiating,

$$\frac{\mathrm{d}S}{\mathrm{d}r} = 4\pi r - \frac{2V}{r^2} = \frac{2(2\pi r^3 - V)}{r^2}.$$

This is negative for $r^3 < V/(2\pi)$ and positive for $r^3 > V/(2\pi)$. So S decreases and then increases, and its least value on $r > 0$ is at

$$r^3 = \frac{V}{2\pi}.$$

This is the global minimum, not just a local one, because it is the only stationary point and the derivative changes sign from $-$ to $+$ there. (You can also see it from the formula: $S \to \infty$ both as $r \to 0^+$ and as $r \to \infty$.)

At this radius $V = 2\pi r^3$, so

$$h = \frac{V}{\pi r^2} = 2r.$$

The best tin is exactly as tall as it is wide. Then $2V/r = 4\pi r^2$, so

$$\begin{aligned} S &= 2\pi r^2 + 4\pi r^2 = 6\pi r^2 = 6\pi \left(\frac{V}{2\pi}\right)^{2/3} \\ &= 3 \cdot 2\pi \cdot (2\pi)^{-2/3} V^{2/3} = 3\,(2\pi)^{1/3} V^{2/3} = 3\,(2\pi V^2)^{1/3}, \end{aligned}$$

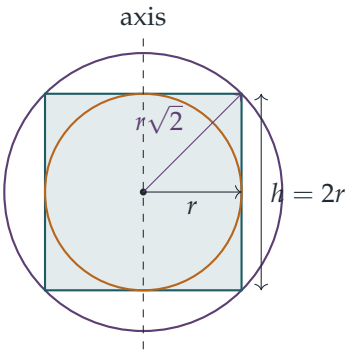
which is the required minimum.

Ask yourself Why work with r rather than h?

Either variable works. Eliminating h is easier, though, because h appears only linearly in both formulae, so the substitution $h = V/(\pi r^2)$ gives a simple function of r. Eliminating r would bring in $\sqrt{V/(\pi h)}$, and the algebra gets messier.

The largest sphere inside the tin

From now on the tin has $h = 2r$. Cut it by a plane through its axis: the cross-section is a $2r \times 2r$ square.



Any sphere inside the tin has radius $\rho \le r$. To see this, project it onto the plane of the base: it gives a disc of radius ρ lying inside the base disc of radius r. Also $2\rho \le h = 2r$, which gives the same bound. The sphere of radius r centred at the centre of the tin (the amber circle in the cross-section) touches both ends and the curved side, so it fits. Using $r^3 = V/(2\pi)$, the largest sphere therefore has volume

$$\tfrac{4}{3}\pi r^3 = \tfrac{4}{3}\pi \cdot \frac{V}{2\pi} = \tfrac{2}{3}V,$$

as required.

The smallest sphere containing the tin

Take a point on the rim of the top end and the diametrically opposite point on the rim of the bottom end. These are opposite corners of the square cross-section, so their

distance apart is

$$\sqrt{(2r)^2 + h^2} = \sqrt{8r^2} = 2\sqrt{2}\,r.$$

Any sphere containing the tin contains both points, so its diameter is at least $2\sqrt{2}\,r$ and its radius at least $\sqrt{2}\,r$.

Ask yourself How do we know a sphere of radius $\sqrt{2}\,r$ is actually big enough?

Centre it at the centre O of the tin. Any point of the tin is at distance $\rho \le r$ from the axis and at height z above or below O with $|z| \le r$. Its distance from O is therefore $\sqrt{\rho^2 + z^2} \le \sqrt{r^2 + r^2} = \sqrt{2}\,r$. So the whole tin lies inside this sphere (the plum circle passes through all four corners of the square).

The smallest sphere into which the tin fits therefore has radius $\sqrt{2}\,r$ and volume

$$\tfrac{4}{3}\pi\left(\sqrt{2}\,r\right)^3 = \tfrac{4}{3}\pi \cdot 2\sqrt{2}\,r^3 = \tfrac{8\sqrt{2}}{3}\pi \cdot \frac{V}{2\pi} = \frac{4\sqrt{2}}{3}\,V.$$

Checks

- With $V = 1$, minimising $2\pi r^2 + 2/r$ by brute force over a fine grid of r gives $S \approx 5.5358$, which agrees with $3(2\pi)^{1/3} \approx 5.5358$.
- The ratio of the two sphere volumes should be $(\sqrt{2})^3 = 2\sqrt{2}$, the cube of the ratio of the radii. Indeed $\frac{4\sqrt{2}}{3}V \big/ \frac{2}{3}V = 2\sqrt{2}$.
- The order makes sense: $\frac{2}{3}V < V < \frac{4\sqrt{2}}{3}V \approx 1.886\,V$, so the inner sphere is smaller than the tin and the outer one larger.

Common slips

- Forgetting one or both ends, writing $S = \pi r^2 + 2\pi rh$ or $S = 2\pi rh$. That gives a different optimal shape.
- Showing $\mathrm{d}S/\mathrm{d}r = 0$ without saying why the stationary point is a minimum.
- Taking the radius of the outer sphere to be r or h, rather than half the space diagonal of the tin.

Beyond STEP

This is constrained optimisation: minimise $S(r,h)$ subject to $V(r,h)$ being fixed. At university you will meet Lagrange multipliers, which handle this without eliminating a variable. The condition $\nabla S = \lambda \nabla V$ gives $h = 2r$ almost at once. The two spheres also illustrate the "inscribed" and "circumscribed" balls of a convex body, which are central in convex geometry. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

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